MAAI 902 · NATURAL LANGUAGE PROCESSING & LLMS · TASK 2

Clinical Note Routing

Leakage-free, explainable specialty classification of medical transcriptions

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THE PROBLEM

Every clinical note has to reach the right specialty

77%

of 1,253 clinicians said EHR documentation made them finish late or work from home, and 74% said it took time away from patients (Healthcare Innovation, 2024).

Task

Given a transcription, assign **every specialty** that should receive it, show **how confident** the system is, and **flag notes** that need a human decision.

One note can belong to several specialties, and a confident wrong answer is worse than asking a person.

WHY NLP

Keyword rules cannot sort clinical language

Free text

Notes average about 400 words, mix headed sections with narrative, and use abbreviations such as EF and ORIF.

Shared words

Words like pain, MRI and biopsy appear in many specialties. Meaning comes from combinations and context.

Tools exist

Clinical language models and biomedical NER are freely available (Alsentzer et al., 2019; Neumann et al., 2019; Li et al., 2023).

RESEARCH GAP

Published results on this dataset disagree

0.28

macro-F1 with a leakage-free split (Barman and Barman, 2026)

0.996

F1 after SMOTE oversampling, test split not described (Pottipati and Polamreddy, 2026)

Leakage is a known cause of results that cannot be reproduced (Kapoor and Narayanan, 2023).

Mixed labels: some labels are document types, such as Discharge Summary, not specialties.

No confidence: studies report accuracy, but routing needs trustworthy scores.

THE DATA

3,813 rows hold only 1,821 unique notes

3,813

rows in the course CSV, 39 labels, no missing values

→

1,821

unique transcriptions after removing duplicates

→

1,751

clean notes and 24 specialties, 1.82 labels per note

Source: MTSamples, a public library of transcription samples where names and dates have been changed or removed (MTSamples, n.d.). Only the **transcription** text is used as model input.

DATA AUDIT

Three findings changed the design

100%

Label leakage

Every keywords value starts with the specialty name. The column is never used.

1,650

Multi-label notes

notes are filed under 2+ specialties. A colonoscopy is Gastroenterology and Surgery.

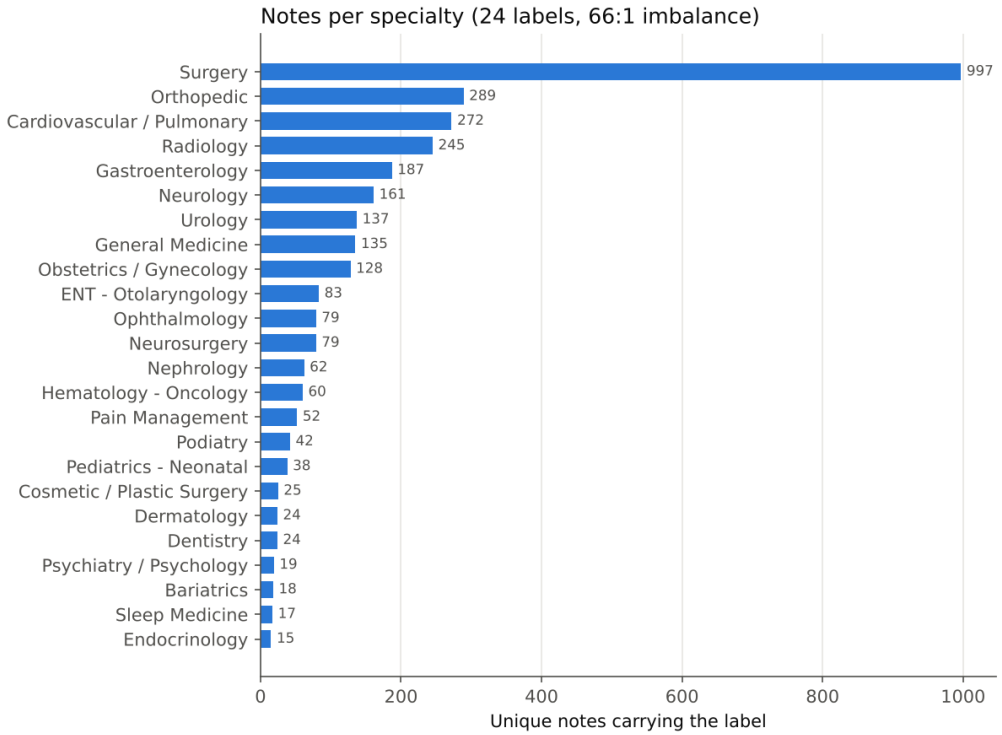
7

Mixed labels

labels are document types, such as Discharge Summary and Letters. They were removed.

EXPLORATORY ANALYSIS

Heavy imbalance and long notes



66 : 1 imbalance

Surgery has 997 notes, Endocrinology 15. Macro-F1 is the main metric so every specialty counts equally.

45% of notes are long

They exceed about 512 BERT tokens (median 356 words), so transformers must split them into windows.

Overlap is structural

Surgery shares 201 notes with Orthopedic; Radiology shares 79 with Neurology.

ARCHITECTURE

From a note to a routing decision

Clinical note
free-text input

→

Clean text
lower case, strip marks

→

TF-IDF
11,017 words

→

Classifiers
24, one per specialty

→

Assign
p ≥ 0.34, plus the top-1

Confidence ≥ 0.70: auto-route
87.5% of notes, 96.7% top-1 accuracy

Below 0.70: human review
12.5% of notes go to a person

Clinical NER
diseases and drugs added to the slip

Every slip also shows the **evidence words** behind the decision: TF-IDF value × model weight.

MODELS AND TUNING

Five models, the same five folds

Model	Why it was included
Prior (always Surgery)	Minimum baseline every model must beat
TF-IDF + Naive Bayes	Classic text baseline; best in Kathirvel and Mookiah (2025)
TF-IDF + linear SVM	Strong linear classifier for sparse text
TF-IDF + logistic regression	Gives probabilities for review and a weight for every word
Bio_ClinicalBERT (chunked)	Clinical pre-training (Alsentzer et al., 2019)

Leakage-free folds: iterative stratification on unique notes (Sechidis, Tsoumakas and Vlahavas, 2011).

Tuning: C in 1–64 × single words or word pairs. Best: C = 64, single words.

BERT: 512-token windows, stride 128, 4 epochs, learning rate 2e-5.

RESULTS · RQ1

Logistic regression wins on every metric

Model	Macro-F1	Micro-F1	Top-1
Prior (always Surgery)	0.030	0.404	0.569
TF-IDF + Naive Bayes	0.545	0.830	0.870
TF-IDF + linear SVM	0.780	0.886	0.925
TF-IDF + logistic regression	0.809	0.892	0.934
Bio_ClinicalBERT (fold 0)	0.544	0.741	0.831
Logistic regression (fold 0)	0.805	0.888	0.931

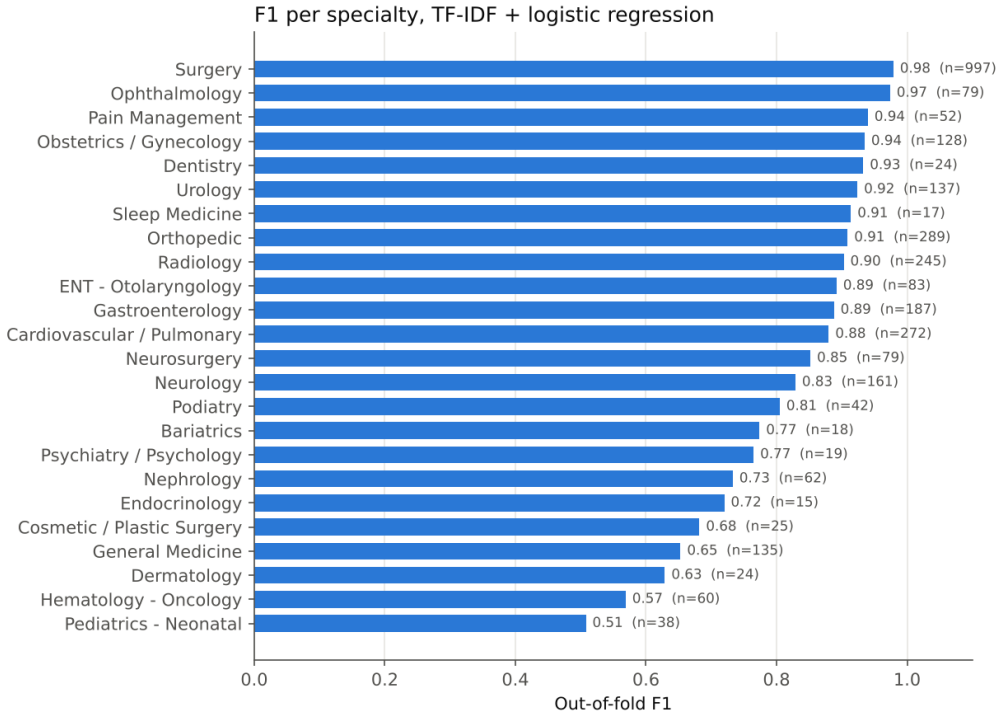
0.809

macro-F1 ± 0.033 over five folds, with 93.4% of notes getting a correct top specialty

BERT scored lower on the same fold, which matches earlier findings that larger models do not always win on small data (Kathirvel and Mookiah, 2025; Dada et al., 2025).

ERROR ANALYSIS

Procedures are easy, patient groups are hard



Strongest

Surgery 0.98, Ophthalmology 0.97, Obstetrics / Gynecology 0.94: clear procedure words.

Weakest

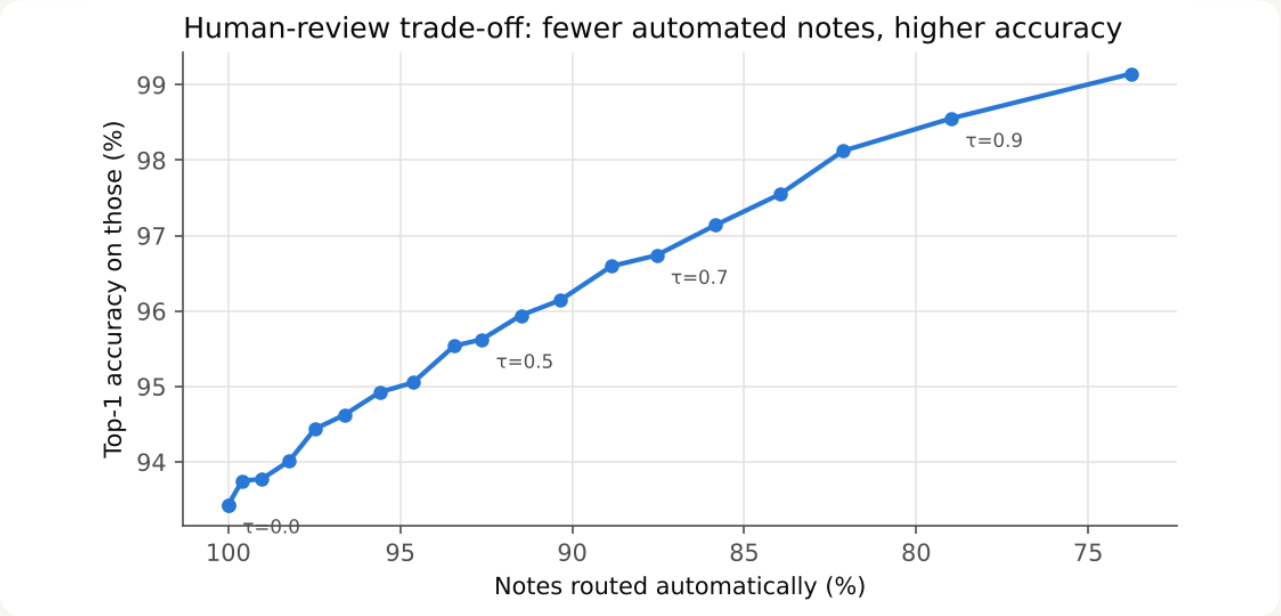
Pediatrics 0.51, Hematology–Oncology 0.57, Dermatology 0.63: defined by patient group or broad scope.

Small is not always weak

Sleep Medicine has 17 notes but reaches 0.91: sleep, apnea and EEG are very specific.

HUMAN REVIEW · RQ3

Trusted confidence makes review safe



τ	Auto-routed	Accuracy
none	100%	93.4%
0.5	92.6%	95.6%
0.7	87.5%	96.7%
0.9	79.0%	98.6%

Calibration error is only 0.039 (Guo et al., 2017). At τ = 0.7, one note in eight goes to a person and errors on the rest fall by half, from 6.6% to 3.3%.

ABLATION · RQ2

Same model, accuracy from 0.10 to 0.81

#	Evaluation method	Acc.	Macro-F1
E0	Raw rows, one label, random split	0.104	0.057
E1	E0 + oversampling before the split	0.807	0.783
E2	E0 + oversampling after the split	0.104	0.071
E3	E2 + keywords column	0.722	0.529
E4	Raw rows, grouped split	0.444	0.202
E5	Multi-label, all 39 labels	top-1 0.946	0.695
E6	Multi-label, 24 labels (this design)	top-1 0.934	0.809

Single-label fails: the test text is in training under another correct label.

Leakage inflates scores (E1, E3), as warned by Kapoor and Narayanan (2023) and Barman and Barman (2026).

The label fix adds real value: 0.695 → 0.809.

RESPONSIBLE AI

Risks and how the design reduces them

Risk	Mitigation
Patient privacy	De-identified data; the model runs locally; no external LLM services
Specialty bias	F1 reported per specialty; class weights; low-confidence notes go to review
Automation bias	Confidence on every slip; adjustable threshold; evidence words to check
Overstated results	Leakage audit, folds on unique notes, protocol ablation, variance reported
Accountability	The system supports people; logs record model version, confidence, decision

The human-review threshold is the central safeguard. Any future LLM summary would be grounded in the note and checked by a person (Van Veen et al., 2024).

PROTOTYPE

Every note gets a routing slip

EXAMPLE · LUMBAR SPINE CT

Route automatically · 99.96% confidence

Send to: Radiology, Orthopedic, Neurology

Source labels: Neurology, Orthopedic, Radiology

Evidence: spine, images, exam, impression, technique

The web demo runs fully in the browser: no note leaves the device.

A slider changes the review threshold live.

A Gradio app adds the diseases and drugs found by NER.

Two Colab notebooks reproduce every result.

CONCLUSION

The data decisions mattered most

RQ1

Without leakage, logistic regression reaches macro-F1 0.809 and 93.4% top-1 accuracy.

RQ2

The evaluation method alone moves accuracy from 0.10 to 0.81; the label fix adds 0.11 macro-F1.

RQ3

Calibrated confidence (ECE 0.039) routes 87.5% of notes at 96.7% accuracy.

Checking for leakage, treating the task as multi-label and separating specialties from specialties from document types mattered more than the choice of model.

LIMITATIONS AND NEXT STEPS

What this study cannot yet show

Limitations

- Sample transcriptions, not real hospital records
- Labels filed by users, not medical experts
- BERT tested on one fold, without tuning
- NER accuracy not measured; English only

Future work

- Tuned BERT on five folds; Clinical-Longformer for whole notes
- Model label links, as label-graph methods do
- Test on approved hospital data with clinicians
- A separate document-type classifier

Key references

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Healthcare Documentation Database

Full Harvard reference list in the report